SOCIOL 723: Social Statistics II

Week 12, Machine Learning: Prediction and Regularization

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Today

The Bias–Variance Tradeoff

Regularization

Trees and Ensembles

★ = essential, you must understand this ★ = for understanding, not examined

A Different Goal ★

Weeks 5–11 asked: *what is the causal effect, and what must be true to recover it?* Today we set that aside entirely.

Prediction versus inference

Inference: we care about a parameter β, its interpretation, and its standard error.
Prediction: we care only about $\hat{Y}$ for units we have not seen. The parameters are means, not ends, and may not even be identified.

- These goals conflict. The model that predicts best is usually not the model whose coefficients you would want to interpret, and vice versa.
- Molina and Garip (2019): sociology has plenty of genuine *prediction* problems (who drops out, which neighbourhoods gentrify, which cases to audit), and separate ML tools for describing heterogeneity.
- Next week we put the two back together. Today, keep them apart.

Bias-Variance

Decomposing Prediction Error ★

Let $Y = f(X) + \epsilon$ with $E[\epsilon \mid X] = 0$ and $\text{Var}(\epsilon \mid X) = \sigma^2$, and let $\hat{f}$ be estimated from a training sample drawn independently of the new observation. At a new point x_0, taking expectations over *both* the training sample and the new outcome:

$$E\left[\left(Y_0 - \hat{f}(x_0)\right)^2\right] = \underbrace{\left(E[\hat{f}(x_0)] - f(x_0)\right)^2}_{\text{bias}^2} + \underbrace{\text{Var}\left(\hat{f}(x_0)\right)}_{\text{variance}} + \underbrace{\sigma^2}_{\text{irreducible}}.$$

- **Bias**: how far the method is wrong on average, across training samples. Rigid models (a straight line) have high bias.
- **Variance**: how much $\hat{f}$ moves if you redraw the training data. Flexible models (a deep tree) have high variance.
- **Irreducible error**: noise. No method beats it.

Decomposing Prediction Error ★ (cont.)

- Every tuning parameter in this course is a dial between the first two terms.
  Penalty size, tree depth, number of neighbours, bandwidth (Week 9!), all the same dial.

Why Unbiasedness Stops Being the Goal ★

Weeks 1–2 prized unbiasedness: Gauss–Markov ranked estimators *within* the unbiased class.

For prediction, that is the wrong criterion

We care about total error. Accepting a little bias to remove a lot of variance lowers MSE. Every method today is deliberately biased.

- OLS with k regressors has variance growing in k: with p/n not small, the variance term swamps everything and OLS predicts badly even though it is unbiased.
- When $p > n$, OLS is not even defined, $\boldsymbol{X}'\boldsymbol{X}$ is singular.
- **In-sample fit is not evidence.** R^2 never falls when you add regressors, so training error is a systematically optimistic estimate of test error. The gap is the *optimism*, and it grows with model complexity.

Estimating Test Error: Cross-Validation ★

Split the data, fit on part, evaluate on the rest.

K-fold cross-validation

Partition the sample into K folds. For $k = 1, \ldots, K$: fit on all folds but k, predict fold k, record the error. Average:

$$CV_K = \frac{1}{K} \sum_{k=1}^{K} \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \left(Y_i - \hat{f}^{(-k)}(X_i) \right)^2.$$

- $K = 5$ or 10 is standard: leave-one-out ($K = n$) has low bias but high variance and is expensive.

Estimating Test Error: Cross-Validation ★ (cont.)

- Everything data-dependent must happen inside the fold, variable selection, standardization, imputation. Selecting predictors on the full sample and then cross-validating is a very common and very serious leak.
- With dependence (panels, clusters, time) split on the independent unit, not on rows. Same rule as the bootstrap in Week 2.

Regularization

Ridge Regression ★

Add a penalty on the squared size of the coefficients:

$$\hat{\boldsymbol{\beta}}^{\text{ridge}} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} (Y_i - X_i'\boldsymbol{\beta})^2 + \lambda \sum_{j=1}^{p} \beta_j^2, \qquad \hat{\boldsymbol{\beta}}^{\text{ridge}} = (\boldsymbol{X}'\boldsymbol{X} + \lambda \boldsymbol{I})^{-1}\boldsymbol{X}'\boldsymbol{y},$$

where, as always with a penalty, $\boldsymbol{X}$ and $\boldsymbol{y}$ are centred and the intercept is left out of both the penalty and the displayed matrix.

- Adding $\lambda \boldsymbol{I}$ makes the matrix invertible even when $p > n$, which is why ridge works where OLS does not exist.
- Coefficients shrink toward zero but never reach it: ridge does not select.
- **Always standardize the predictors first**; the penalty is not scale-invariant, and the intercept is not penalized.

Ridge Regression ★ (cont.)

- Ridge shines under dense signals: many small effects, and correlated predictors whose coefficients OLS cannot separate.
- As $\lambda \to 0$ we recover OLS; as $\lambda \to \infty$ all slopes go to zero and the fit collapses to the sample mean of Y. The dial is the bias–variance dial.

Lasso ★

Penalize the *absolute* size instead:

$$\hat{\boldsymbol{\beta}}^{\text{lasso}} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \left(Y_i - X_i'\boldsymbol{\beta}\right)^2 + \lambda \sum_{j=1}^{p} |\beta_j|.$$

- The ℓ_1 penalty sets coefficients exactly to zero, so lasso performs variable selection and estimation simultaneously.
- Geometrically: the ℓ_1 constraint region is a diamond with corners on the axes, and the contours of the sum of squares typically first touch it at a corner. The ℓ_2 region is a ball with no corners, hence no exact zeros.
- Lasso suits approximate sparsity: a few predictors matter, the rest are approximately irrelevant.
- **Elastic net** mixes them, $\lambda\left(\alpha \sum |\beta_j| + \frac{1-\alpha}{2} \sum \beta_j^2\right)$, which handles groups of correlated predictors better than pure lasso.

Choosing λ, and a Warning About Interpretation ★

- **Cross-validation** over a grid of λ: pick the minimizer, or the "one standard error" rule, the largest λ within one standard error of the minimum, which buys a simpler model at almost no predictive cost.
- There are also **theoretically derived** penalties that set λ from the model rather than by cross-validation. They matter when the lasso is a *step inside* an inference procedure rather than the final answer.

Lasso selection is not a variable-importance ranking

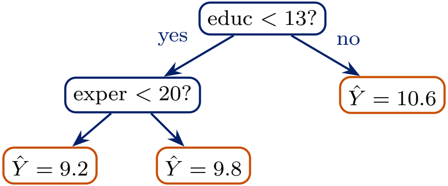
Among correlated predictors, lasso picks one essentially arbitrarily and zeroes the rest. Re-run on a bootstrap sample and you may get a different set. Do not interpret the selected set as "the variables that matter", and do not report post-lasso p-values from a naive refit, the selection invalidates them. Valid post-selection inference is exactly what Week 13 is for.

Trees

Regression Trees ★

Ridge and lasso control complexity, but both still start from an *additive linear* rule: you choose the variables, and an interaction exists only if you type it. Trees are the first method we meet that finds nonlinearities and interactions on its own.

Recursively split the covariate space to minimize within-node sum of squares; the prediction in a leaf is the mean of the training outcomes there.



- **Strengths**: automatic interactions and nonlinearity; invariant to monotone transformations; handles mixed variable types; readable.

Regression Trees ★ (cont.)

- Weaknesses: high variance, a small change in the data can change the first split and hence everything below it. A single deep tree overfits badly; a shallow one underfits.
- Pruning with a complexity penalty helps but does not solve it. The real fix is to average many trees.

Bagging, Random Forests, Boosting ★

- **Bagging**: fit trees on bootstrap resamples and average. Averaging B nearly independent estimates cuts variance by roughly $1/B$ without adding bias. But bootstrap trees are *correlated*, they all split on the same dominant predictor first, which caps the gain.
- **Random forests**: bag, but at each split consider only a random subset of m predictors. Decorrelating the trees is what makes the averaging work. m is the main tuning parameter, alongside node size. Out-of-bag error is a free cross-validation estimate.
- **Boosting**: fit trees *sequentially*, each on the residuals of the current ensemble, adding a shrunken version: $\hat{f} \leftarrow \hat{f} + \eta\,\hat{t}$. Attacks *bias* rather than variance, and with small η and many trees is usually the strongest tabular predictor available.

Bagging and random forests reduce variance; boosting reduces bias. That is the cleanest way to keep them straight.

What ML Buys, and What It Does Not ★

Buys

- Prediction with many predictors, including $p > n$.
- Automatic nonlinearity and interactions, no functional form to guess.
- Honest out-of-sample performance estimates.

Does not buy

- Causal interpretation. A variable importance score is not an effect.
- Valid standard errors for the coefficients it selects.
- Extrapolation: trees do return a prediction for out-of-range inputs, but only the boundary leaf's constant, so they cannot extend a trend.
- Freedom from the identification problems of Weeks 5–11.

What ML Buys, and What It Does Not ★ (cont.)

The genuine sociological uses

Prediction as the goal (screening, triage, forecasting); *measurement* (constructing variables from text or images); *describing heterogeneity*; and estimating nuisance functions inside a causal estimator, which is next week.

Looking Ahead

- Every method today trades bias for variance, tuned by cross-validation.
- That trade is exactly why we cannot drop these estimates into a regression and read off a coefficient: they are *deliberately biased* and converge slowly.
- **Next week**: what has to change before an estimate this biased can be used as an ingredient in a causal estimator, and what it buys us when it can.

Problem Set 5

Assigned today, due Monday November 23. Covers Weeks 10–11.

Reading

ISL Chapters 5, 6, and 8; Molina and Garip (2019). Optional: ISL Chapter 10 (deep learning); Kleinberg et al. (2015) on prediction policy problems.

References

- James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani. 2021. *An Introduction to Statistical Learning*, 2nd ed. Springer. [Ch. 5, 6, 8]
- Kleinberg, Jon, Jens Ludwig, Sendhil Mullainathan, and Ziad Obermeyer. 2015. "Prediction Policy Problems." *American Economic Review: Papers & Proceedings* 105(5): 491–495.
- Molina, Mario and Filiz Garip. 2019. "Machine Learning for Sociology." *Annual Review of Sociology* 45(1): 27–45.
- Hastie, Trevor, Robert Tibshirani, and Jerome Friedman. 2009. *The Elements of Statistical Learning*, 2nd ed. Springer.
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